Design and Analysis of Algorithms

MCQ's

Q) what is the maximum no. of divisions made by Euclid's algorithm for the inputs m & n for any values in the range $1 < m < n \leq 10$?

A: Given range : $1 < m < n \leq 10$

$$F_n = F_{n-1} + F_{n-2}$$

$$F_1 = F_2 = 1 \quad \text{(Base case)}$$

$F_3 = F_2 + F_1 = 1+1 = 2$; $F_6 = F_5 + F_4 = 5+3 = 8$

$F_4 = F_3 + F_2 = 2+1 = 3$; $F_7 = F_6 + F_5 = 8+5 = 13$ ×

$F_5 = F_4 + F_3 = 3+2 = 5$ ↳ 13 out of range

$\therefore m = 8, n = 5$

now, $gcd(m,n) = gcd(8,5)$

$= gcd(5, 8 \bmod 5) \longrightarrow 1$

$= gcd(5,3)$

$= gcd(3, 5 \bmod 3) \longrightarrow 1$

$= gcd(3,2)$

$= gcd(2, 3 \bmod 2) \longrightarrow 1$

$= gcd(2,1)$

$= gcd(1, 2 \bmod 1) \longrightarrow 1$

$= gcd(1,0)$

$= \underline{1} \longrightarrow 1$

$\underline{\underline{5}}$

$\therefore$ Maximum no. of divisions = $\underline{5}$

Q) what will be the output list after completing the first pass of bubble sort on the i/p array.

i/p array : 32 51 27 85 66 23 13 57

A:

Q) Define and explain different asymptotic notations along with example for each.

A: There are mainly three asymptotic notations:

1. Big-Oh notation
2. Ω - notation (Omega)
3. θ - notation (theta)

1. Big-Oh notation

Type: MCQ

Q1. What is the maximum number of divisions made by Euclid's algorithm for the inputs *m* & *n* for any values in the range $1 < m < n \leq 10$? (0.5)

1. ** 5
2. 2
3. 6
4. 3

Q2. Identify the pair of functions in which the first function's order of growth is higher than the second function: (0.5)

1. $n(n + 1)$ and $2000n^2$
2. $(n - 1)!$ and $n!$
3. 2^{n-1} and 2^n
4. ** $log^2{}_2(n)$ and $log_2(n^2)$

Q3. What will be the output list after completing the first pass of bubble sort on the input array 32, 51, 27, 85, 66, 23, 13, 57 ? (0.5)

1. 23, 13, 27, 33, 51, 57, 66, 85
2. 32, 51, 27, 66, 23, 13, 57, 85
3. 27, 33, 51, 23, 13, 57, 66, 85
4. ** 32, 27, 51, 66, 23, 13, 57, 85

Q4. Which of the following can be stated as the problem of finding the shortest Hamiltonian Circuit of the graph? (0.5)

1. Assignment problem
2. **Traveling Salesman problem
3. Breadth first search
4. Depth first search

Q5. How many character comparisons are made by the brute force algorithm in searching for the pattern **RRRRS** in the text

RRRRRRRRRRRRRRRRRRRRRRRRS

(0.5)

1. ** 105
2. 100
3. 21
4. 25

Q6. In what order should we insert the elements {1, 2, 3, 4, 5, 6, 7} into an empty AVL tree so that we don't have to perform any rotations on it?

(0.5)

1. 4,2,1,6,3,5,7
2. 4,2,6,1,3,5,7
3. **4,2,1,6,3,7,5
4. 4,1,2,6,3,7,5

Q7. Which of the following stable sorting algorithm takes the least time when applied to an almost sorted array? (0.5)

1. Quick sort
2. Insertion sort

3. Selection sort
4. ** Merge sort

Q8. Consider the problem of searching an element (x) in an array arr[] of size (n). The problem can be solved in (O(log n)) time if

I.Array is sorted

II.Array is sorted and rotated by (k), where (k) is given and k <= n

III. Array is not sorted (0.5)

1. I only
2. **I and II only
3. I ,II and III only
4. None of the mentioned

Q9. In the context of a extended Binary tree (0.5)

I. The extension of the empty binary tree is a single external node.

II. The number of internal nodes is always 1 more than the number of external nodes.

1. Both are True
2. Both are False
3. **I – True, II – False
4. I – False,II – True

Q10. When a graph is traversed using BFS method, if a new unvisited vertex v is reached for the first time from a current vertex say u, then the edge (u,v) is called as ___________ (0.5)

1. ** Tree Edge
2. Cross Edge
3. Back Edge
4. Solid Edge

11)

There are mainly three asymptotic notations:

1. Big-O notation
2. Omega notation
3. Theta notation

Big-O notation: A function t (n) is said to be in O(g(n)), denoted t (n) ∈ O(g(n)),if t (n) is bounded above by some constant multiple of g(n) for all large n, i.e., if there exist some positive constant c and some nonnegative integer n0 such that **0.5 Marks**

$$t(n) \le cg(n) \text{ for all } n \ge n0.$$

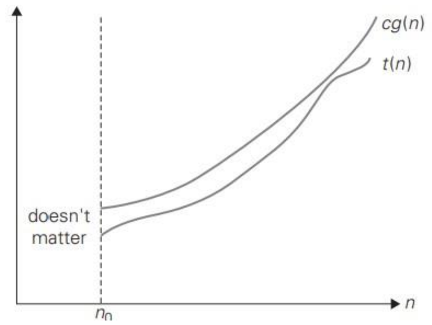


Diagram for θ-notation

Example: 100n + 5 ≤ 100n + n (for all n ≥ 5) = 101n ≤ 101n2 **0.5 Marks**

Ω (Omega) notation : A function t (n) is said to be in Ω (g(n)), denoted t (n) ∈ Ω (g(n)), if t (n) is bounded below by some positive constant multiple of g(n) for all large n, i.e., if there exist some positive constant c and some nonnegative integer n0 such that **0.5 Marks**

t (n) ≥ cg(n) for all n ≥ n0.

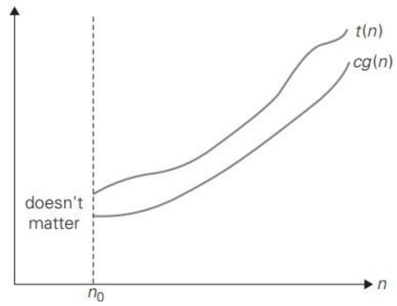
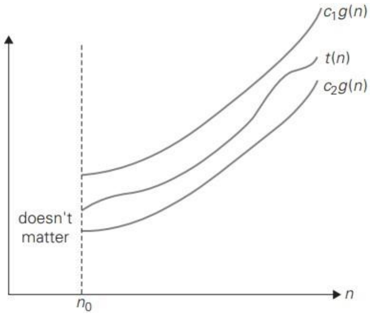


Diagram for Big-O

Example :n3 ∈ (n2): n3 ≥ n2 for all n ≥ 0 **0.5 Marks**

Θ (Theta) notation : A function t (n) is said to be in Θ (g(n)), denoted t (n) ∈ Θ (g(n)), if t (n) is bounded both above and below by some positive constant multiples of g(n) for all large n, i.e., if there exist some positive constants c1 and c2 and some nonnegative integer n0 such that



$c_2g(n) \le t(n) \le c_1g(n)$ for all $n \ge n_0$.
0.5 Marks

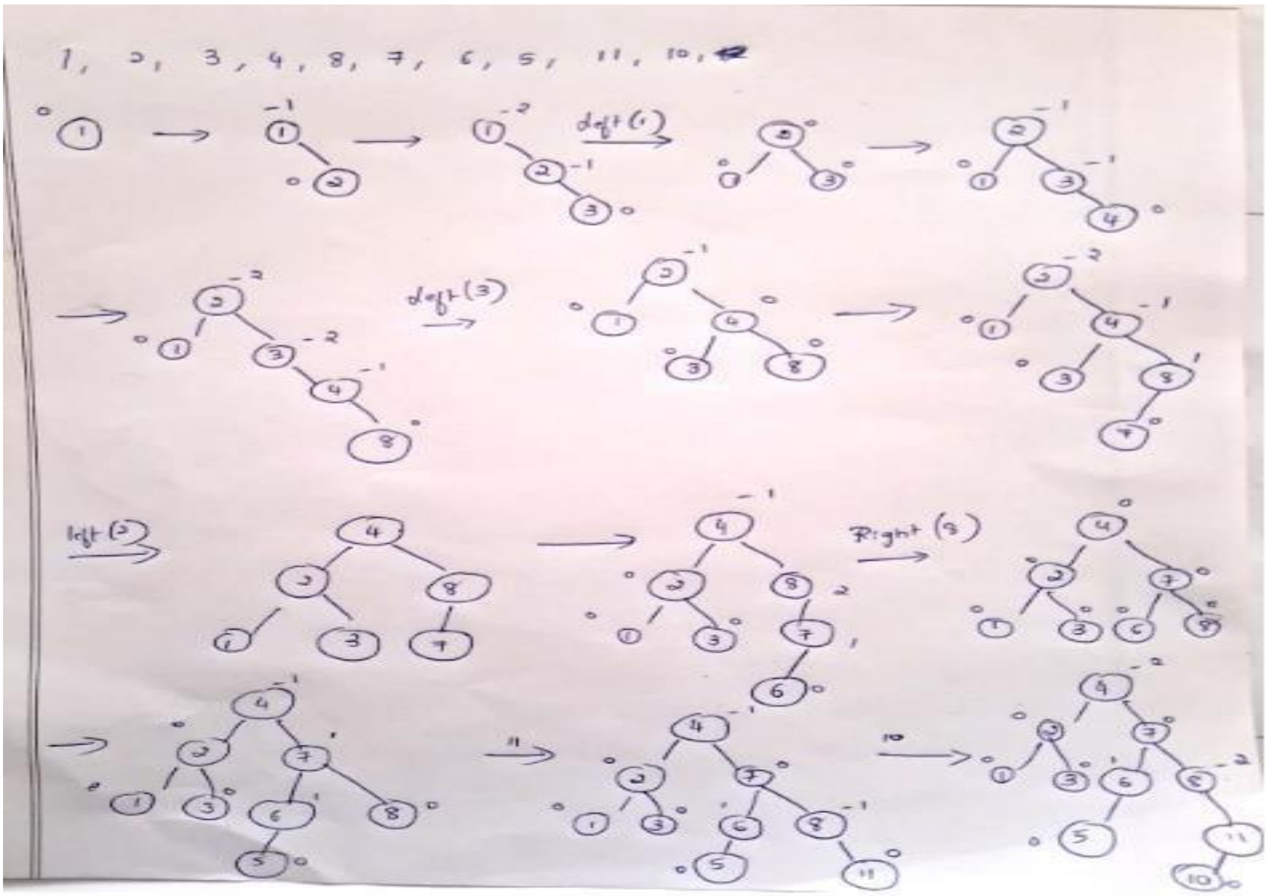
Example :

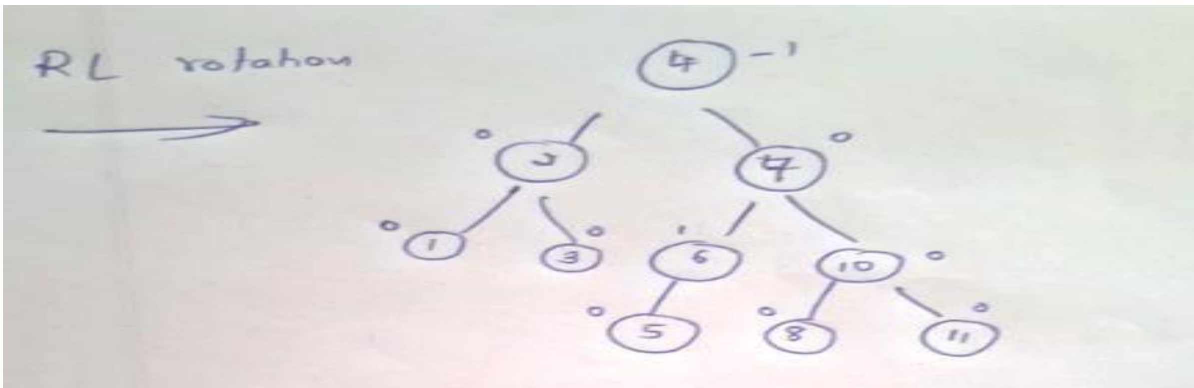
0.5 Marks

$$\frac{1}{2}n(n-1) = \frac{1}{2}n^2 - \frac{1}{2}n \le \frac{1}{2}n^2 \quad \text{for all } n \ge 0.$$

Note: marks have giving which has closer meaning to the definition and one who not written the map.

12)





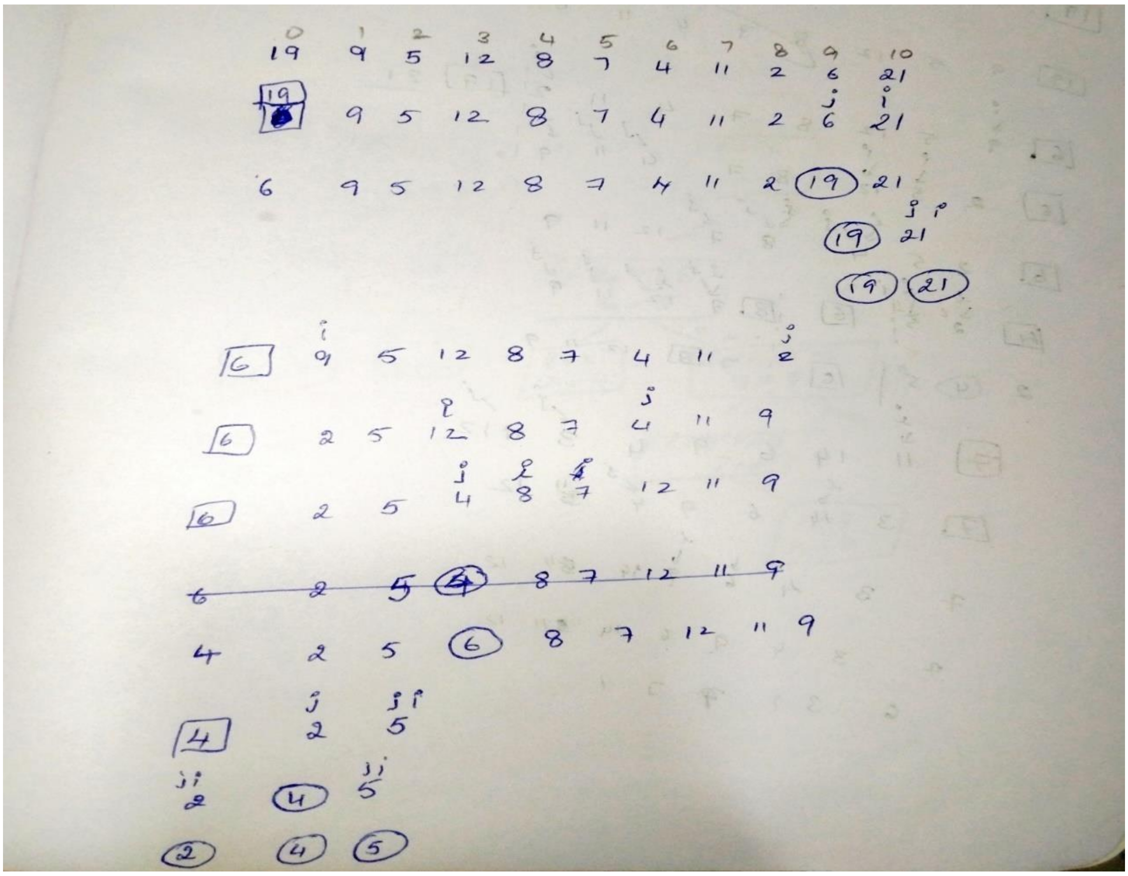
13) Apply quicksort to sort the list: 19, 9, 5, 12, 8, 7, 4, 11, 2, 6, 21 in ascending order. Draw the tree of the recursive calls made.

If any other algorithm other than Hoare is used for partition, then the pseudocode of the algorithm must be written.

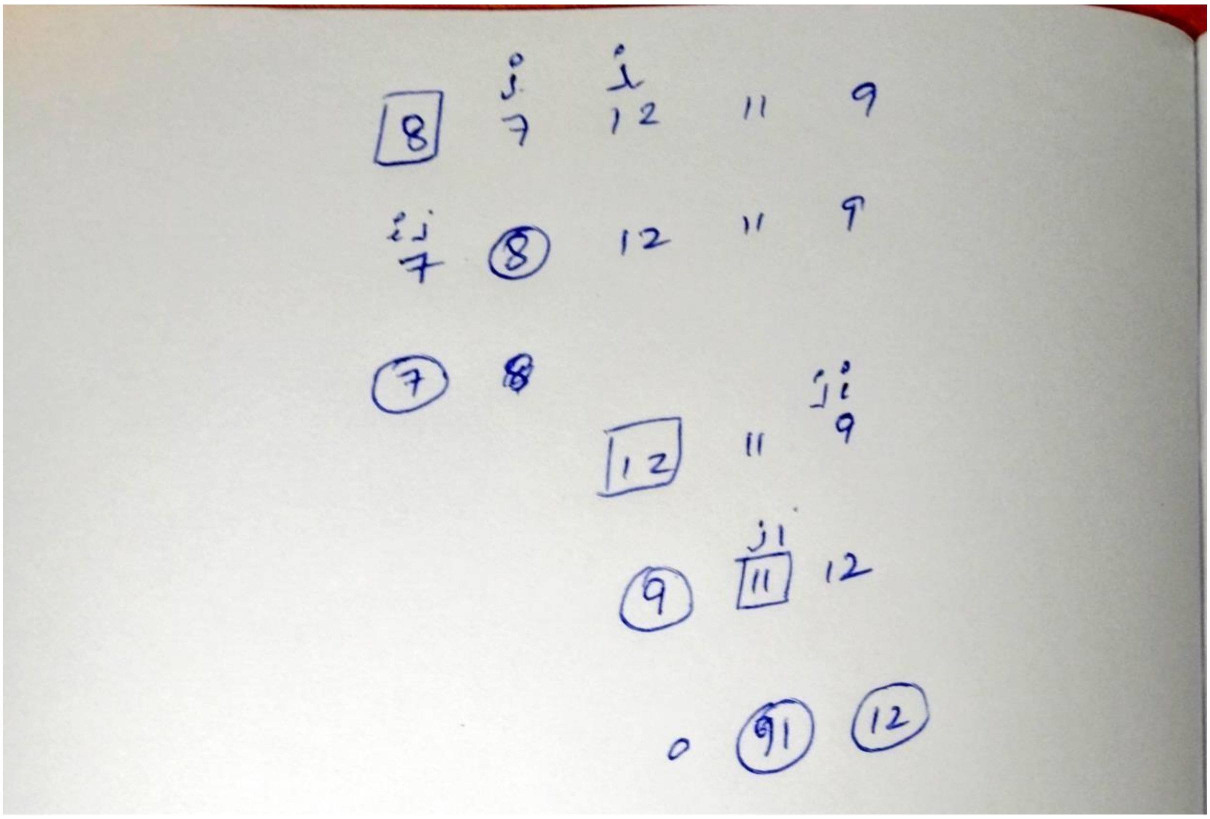
All the steps need to be written clearly.

Refer to the textbook for the format of the tree of the recursive calls.

✞ All the split points needs to be clearly written

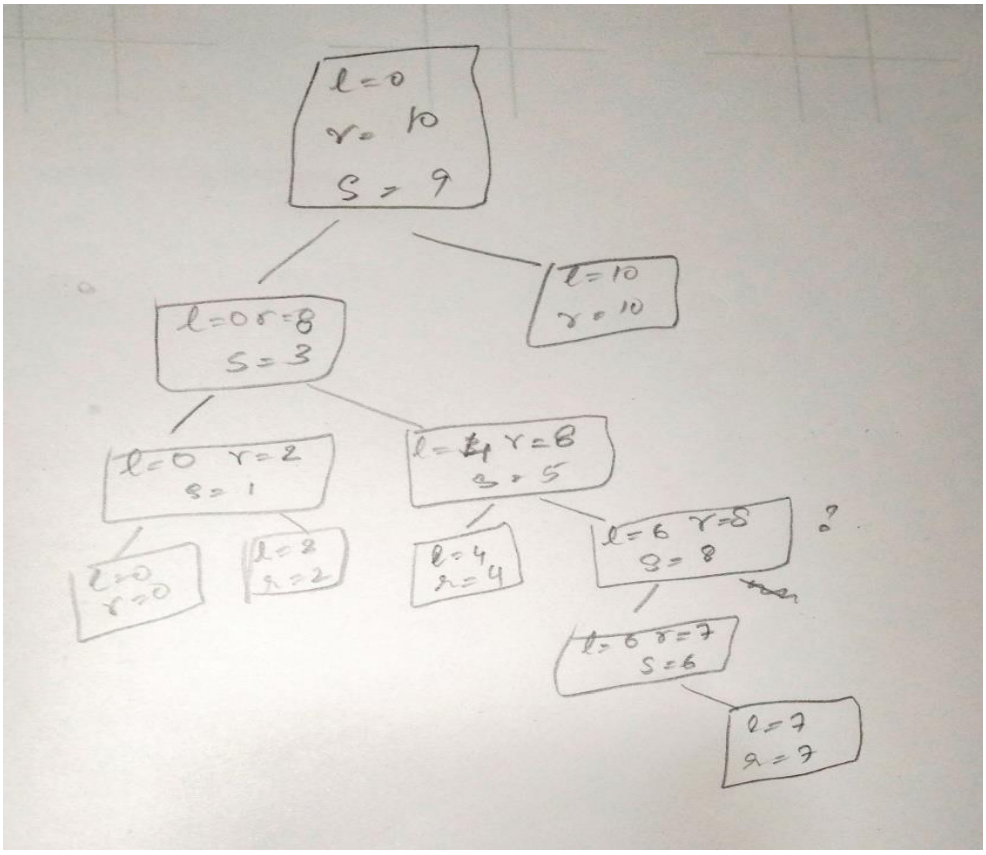


1.5



8 7 12 11 9
7 8 12 11 9
7 8
12 11 9
9 11 12
0 91 12

1.5M



l=0
r= 10
S = 9
l=0 r=8
S=3
l=10
r=10
l=0 r=2
s=1
l=4 r=8
s=5
l=0
r=0
l=2
r=2
l=4
r=4
l=6 r=8
S=8
?
l=6 r=7
S=6
l=7
r=7

1M

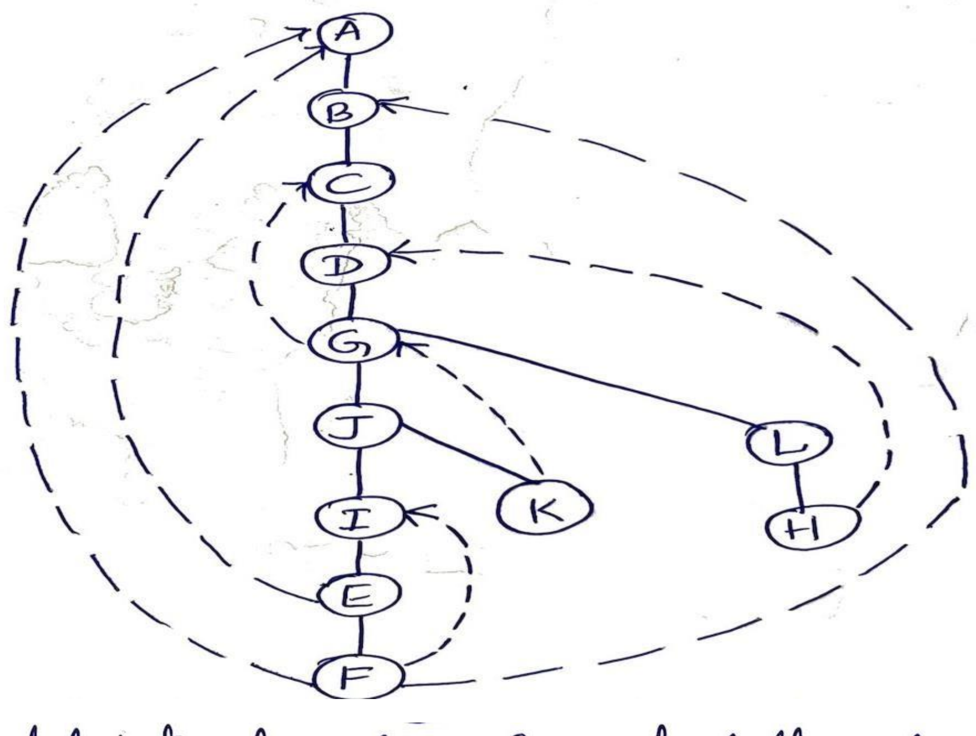
Q14) Push order of vertices - 1 M

A B C D G J I E F K L H

Pop order of vertices - 1 M

F E I K J H L G D C B A

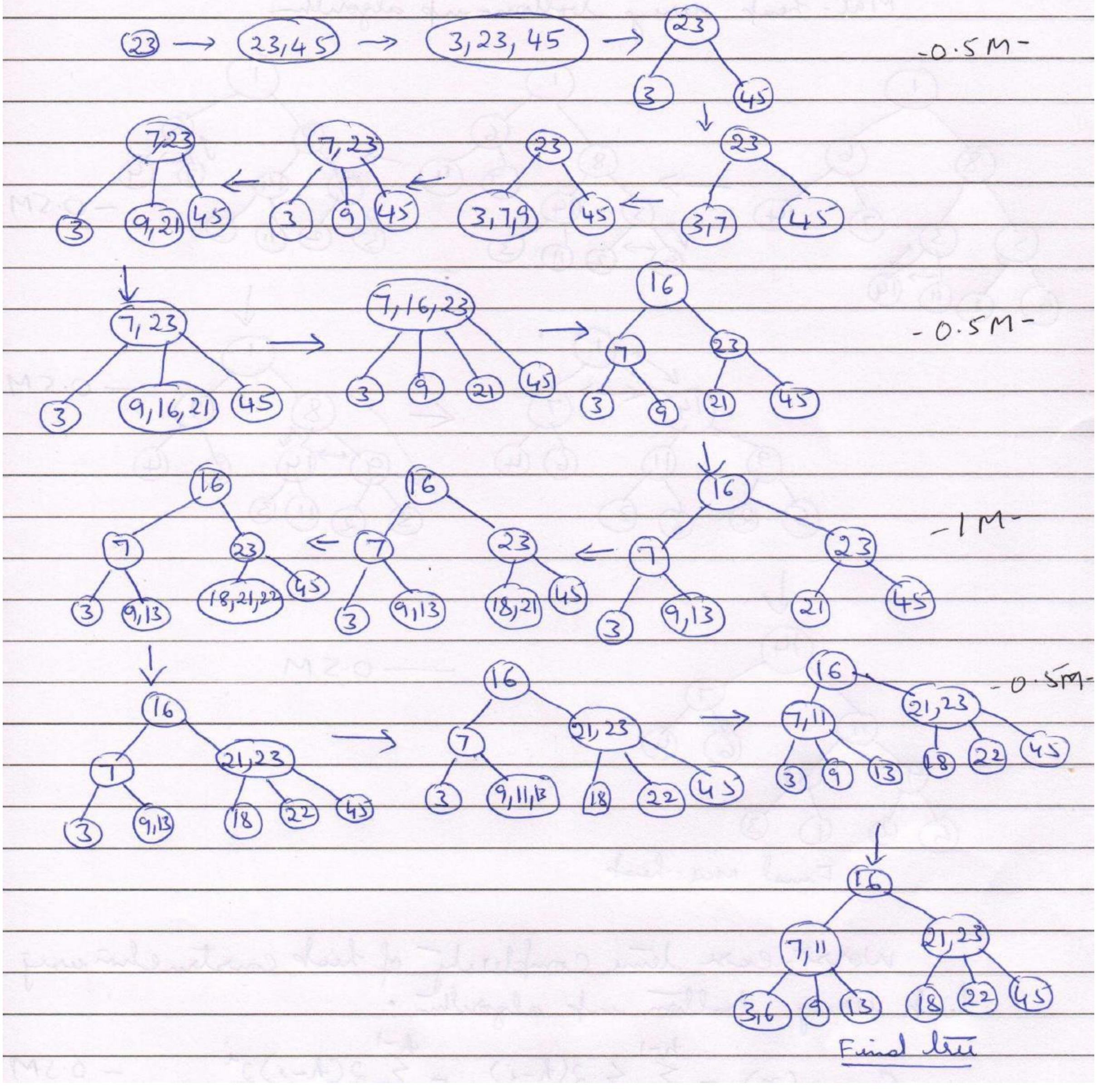
DFS TREE - 1 M



If alphabetical order is not followed marks have been deducted. Whichever part of answer is missing marks are deducted accordingly.

15) 2-3 Tree

23, 45, 3, 7, 9, 21, 16, 13, 18, 22, 11, 6

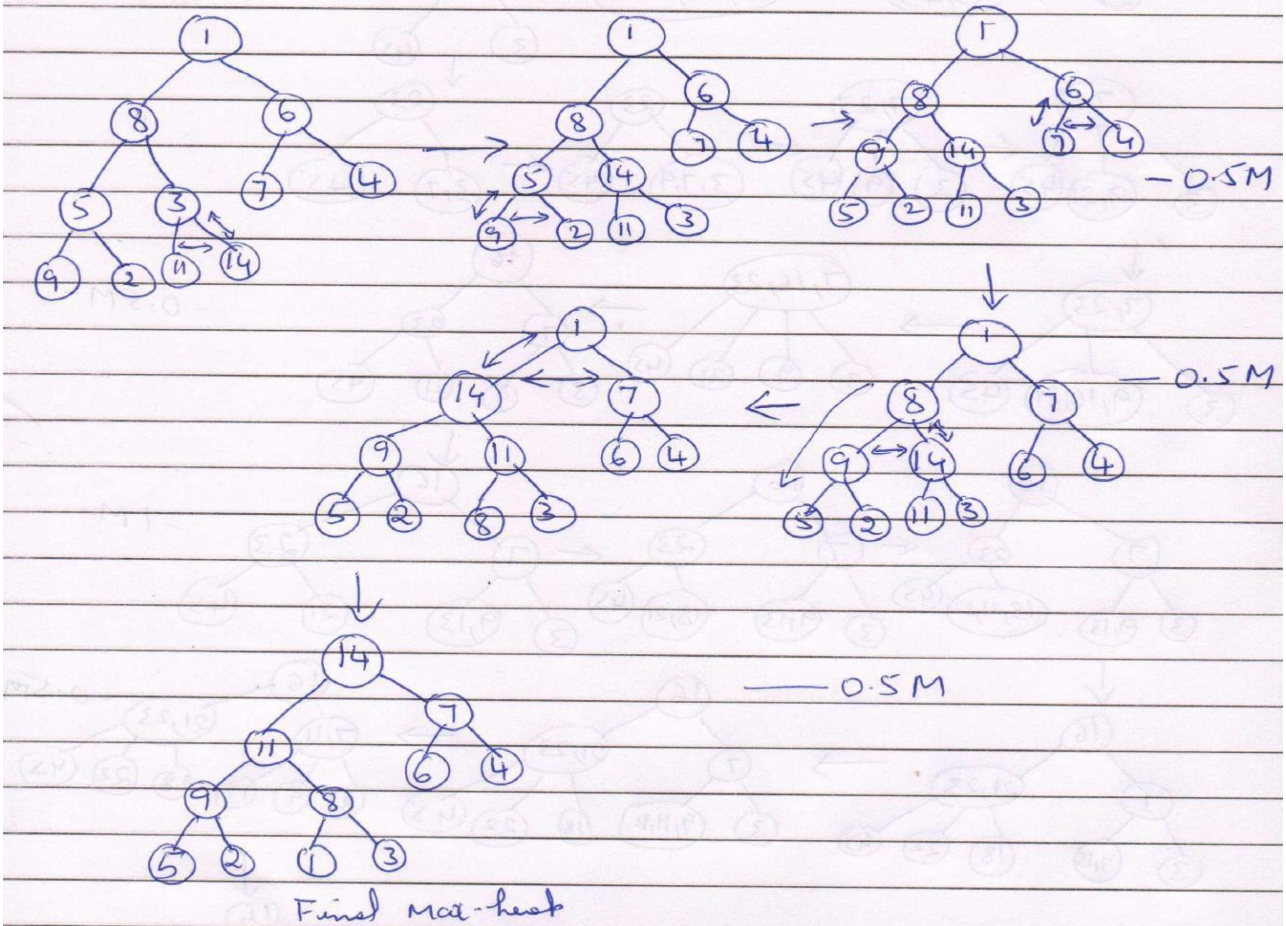


The maximum number of comparisons in the full tree of height 3 is = 8 [Every level 2 comparisons × 4 = 8]

-0.5M-

16] 1, 8, 6, 5, 3, 7, 4, 9, 2, 11, 14

Max-heap using bottom-up algorithm



Final Max-heap

Worst-case time complexity of heap construction using heap using bottom up algorithm:

$$C_{worst}(n) = \sum_{i=0}^{h-1} \sum_{\text{level } i \text{ keys}} 2(h-i) = \sum_{i=0}^{h-1} 2(h-i)2^i \qquad - 0.5M$$

$$= 2\left[\sum_{i=0}^{h-1} h\,2^i - \sum_{i=0}^{h-1} i\cdot 2^i\right]$$

$$= 2\left[h \sum_{i=0}^{h-1} 2^i - \sum_{i=0}^{h-1} i \cdot 2^i\right]$$

$$= 2\left[h(2^h - 1) - [(h-2)2^h + 2]\right]$$

$$= 2\left[h(2^h-1) - h2^h + 2 \cdot 2^h - 2\right]$$

$$= 2\left[\cancel{h \cdot 2^h} - h - \cancel{h \cdot 2^h} + 2 \cdot 2^h - 2\right]$$

$$= 2\left[-h + 2 \cdot 2^h - 2\right]$$

0.5M

$$= 2\left[-h + 2^{h+1} - 2\right]$$

$$= 2\left[-h - 1 + \underbrace{2^{h+1} - 1}_{\rightarrow n}\right]$$

$$\left[\because n = 2^{h+1} - 1\right]$$

$$= 2\left[-\log_2(n+1) + n\right]$$

$$= 2\left[n - \log_2(n+1)\right]$$

$$C_{worst}(n) \in O(n)$$

0.5M

17) Bad symbol shift table.

R	I	M	J	others
5	1	3	2	6

{0.5 M}

Good Suffix shift table

K		
1	RIMJI<u>M</u>	6
2	RIMJ<u>IM</u>	3
3	RIM<u>JIM</u>	6
4	RI<u>MJIM</u>	6
5	R<u>IMJIM</u>	6

{0.5 M}

THE-MUSIC-SOUNDS-RIMJIMSE

RIMJIM
(K=0, d_1=6)

RIMJIM
(K=0, d_1=6)

RIMJIM
(K=0 d_1=5)

RIMJIM

[0.5 M]

No. of comparisons = 1+1+1+6 = 09 [0.5M]

Q18)

Input: b, c, d, c, b, a, a, b, c, a, b

Ans)

Frequencies

a	b	c	d
3	4	3	1

Distribution values

a	b	c	d
3	7	10	11

} 0.5

Ans)

A[10] =	D[a]	D[b]	D[c]	D[d]	S[0]	S[1]	S[2]	S[3]	S[4]	S[5]	S[6]	S[7]	S[8]	S[9]	S[10]
b	3	7	10	11							b				
a	3	6	10	11			a								
c	2	6	10	11										c	
b	2	6	9	11						b					
a	2	5	9	11		a									
a	1	5	9	11	a										
b	0	5	9	11					b						
c	0	4	9	11									c		
d	0	4	8	11											d
c	0	4	8	10								c			
b	0	4	7	10				b							
					a	a	a	b	b	b	b	c	c	c	d

1.5 M

11.

Right linear Grammar	Left Linear Gramar
S→ aaaA	S→Abbb
A→aA|bbB	A→Ab|Baa
B→ b|bB	B→ Ba|λ

(any correct answer 1+1)

12.

1. Assume that the given language is regular
2. Give some number P>0
3. Pick a string w€L such that |w|>=P

   Let w= $a^p b^{p+1}$ since na(w)<nb(w)
4. Given a partition of w= xyz such that |xy|<=p and |y|>0

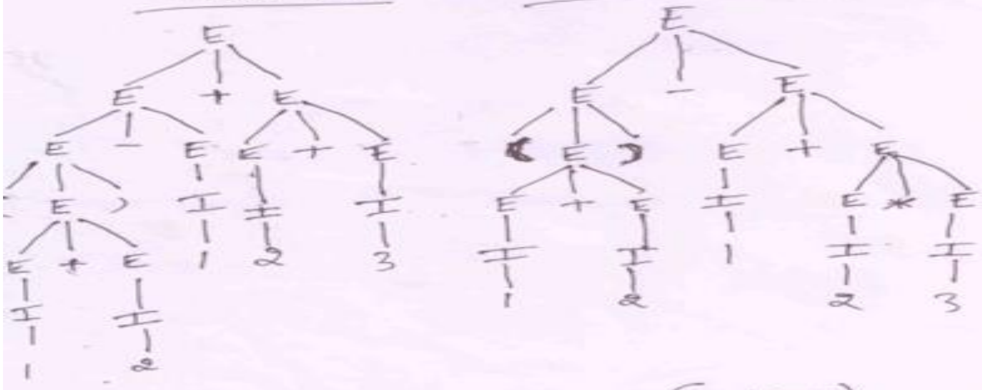
   Let y=a^tt<=p then w= x(empt)y(a)x(b)
5. pick any value of I >=0 and pump w in terms of $xy^i x$ then new string w=$a^{p+t*i}b^{p+1}$

   As the w contains na(w)>nb(w) which does not belong to the given L
6. In contradiction to the assumption the language is no more a regular language.

((steps 1&2→0.5 + steps 3&4 →0.5 + step 5 →0.5 + step 6 →0.5=2))

13) A context free grammar G is said to be ambiguous if there exists some w∈ L(G) that has at least two distinct trees. Also, ambiguity implies the existence of two or more leftmost or rightmost derivations. (1M) w=(1+2)-1+2*3

First Tree Second Tree



(1M) (1M)

Total - 3M

Q: 14) Convert the Grammar(G)=({S, A, B,C,D},{0,1,2},S,P) with the productions

S →0A|0BB|2D , A→00A| λ , B→1B|11C| λ , C→B , D→012DB, E→12E|012 into Chomsky Normal Form.

To convert given grammar into Chomsky Normal Form

$S \rightarrow 0A \mid 0BB \mid 2D$		Remove	$S \rightarrow 0 \mid 0BB \mid 0B \mid 0A \mid 2D$
$A \rightarrow 00A \mid \lambda$		$\Longrightarrow$	$A \rightarrow 00A \mid 00$
$B \rightarrow 1B \mid 11C \mid \lambda$		λ-productions	$B \rightarrow 1 \mid 1B \mid 11C$
$C \rightarrow B$			$C \rightarrow B$
$D \rightarrow 012DB$			$D \rightarrow 012D \mid 012DB$
$E \rightarrow 12E \mid 012$			$E \rightarrow 12E \mid 012$

Remove unit productions

$C \rightarrow B$

$S \rightarrow 0 \mid 0BB \mid 0B \mid 0A \mid 2D$, $A \rightarrow 00A \mid 00$, $B \rightarrow 1 \mid 1B \mid 11C$, $C \rightarrow 1 \mid 1B \mid 11C$,
$D \rightarrow 012D \mid 012DB$, $E \rightarrow 12E \mid 012$

Remove useless productions.

$S \rightarrow 0 \mid 0BB \mid 0B \mid 0A$, $A \rightarrow 00A \mid 00$, $B \rightarrow 1 \mid 1B \mid 11C$, $C \rightarrow 1 \mid 1B \mid 11C$

To convert to Chomsky.

Let $F \rightarrow 0$, $G \rightarrow 1$

$S \rightarrow 0 \mid FBB \mid FB \mid FA$, $A \rightarrow FFA \mid FF$, $B \rightarrow 1 \mid GB \mid GGC$,
$C \rightarrow 1 \mid GB \mid GGC$, $F \rightarrow 0$, $G \rightarrow 1$

Let $H \rightarrow GC$, $I \rightarrow BB$, $J \rightarrow FA$.

$S \rightarrow 0 \mid FI \mid FB \mid FA$, $A \rightarrow FJ \mid FF$, $B \rightarrow 1 \mid GB \mid GH$,
$C \rightarrow 1 \mid GB \mid GH$, $F \rightarrow 0$, $G \rightarrow 1$, $H \rightarrow GC$, $I \rightarrow BB$, $J \rightarrow FA$

$S \rightarrow 0 \mid FI \mid FB \mid FA$	$F \rightarrow 0$
$A \rightarrow FJ \mid FF$	$G \rightarrow 1$
$B \rightarrow 1 \mid GB \mid GH$	$H \rightarrow GC$
$C \rightarrow 1 \mid GB \mid GH$	$I \rightarrow BB$
	$J \rightarrow FA$

Removal of Null production -1M + Removal of Unit Production-1M +
Removal of Useless Production and conversion to CNF 1M

Type: MCQ

Q1. What is the minimum number of states required in a DFA to recognize all the strings not ending with '101' on ∑= (0,1)? (0.5)

1. 1
2. 2
3. 3
4. ** 4

Q2. Identify a regular expression(RE) which generates the language L (r)= $\{a^{2n}b^{2m+1} | n, m\geq 0\}$ (0.5)

1. a*b*b
2. a(aa)*(bb)*b
3. ** (aa)*(bb)*b
4. (ab)*(ba)*bb

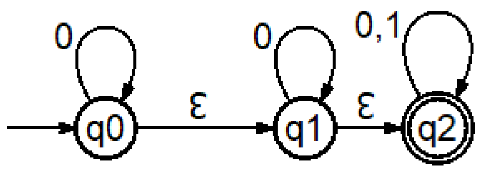
Q3. The minimum states required in NFA to accept all the strings on input (0, 1) where the string starts and with ends with 0. (0.5)

1. 2
2. ** 3
3. 4
4. 5

Q4. Which of the following statements is true about a Deterministic Finite Automaton (DFA)?. (0.5)

1. ** It can recognize all regular languages
2. It can recognize non-regular languages
3. It cannot have more than one final state
4. It can have more than one transition on same input symbol from any state

Q5. Identify the states reachable by $\delta(q_0,0)$ for a below given NFA, where is same as λ? (0.5)



1. ** {q0,q1,q2}

2. {q0,q2}
3. {q1,q2}
4. {q2}

Q6. Which among the following rules are TRUE with respect to identities of the regular expressions?

i. $r + \phi = \phi$

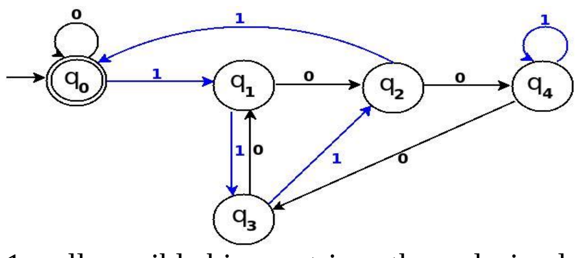
ii. $r \cdot \phi = \phi$

iii. $\phi^* = \lambda$

iv. $\phi^+ = \lambda$ (0.5)

1. ii and iv
2. i and iv
3. **ii and iii
4. i and iii

Q7. ____________ is the language accepted by following DFA. (0.5)



1. all possible binary strings those decimal equivalent is divisible by 4
2. all possible binary strings those decimal equivalent is divisible by 6
3. **all possible binary strings those decimal equivalent is divisible by 5
4. all possible binary strings those decimal equivalent is divisible by 7

Q8. What language does the productions of the given grammar below generate? (0.5)

$S \rightarrow 0A \qquad A \rightarrow 1S \qquad S \rightarrow \lambda$

1. $L(G) = \{0^n 1^m \mid n \geq 1, m \geq 1\}$
2. $L(G) = \{(10)^n \mid n \geq 0\}$
3 . **$L(G) = \{(01)^n \mid n \geq 0\}$
4. $L(G) = \{0^n 1^m \mid n \geq 0, m \geq 0\}$

Q9. How many strings of length less than 4 contains the language described by the regular expression (x+y)*y(a+ab)*?

1. ** 12
2. 10
3. 7
4. 14

Q10. State which is true while converting from NFA to DFA

i. DFA contains has same number of states as that of NFA every time
ii. Number of final states in DFA is equal to final states in NFA
iii. NFA initial state with λ transition to final state, then DFA initial state is also marked as final state

1. i,ii
2. ii,iii
3. only ii
4. ** only iii

Type: DES

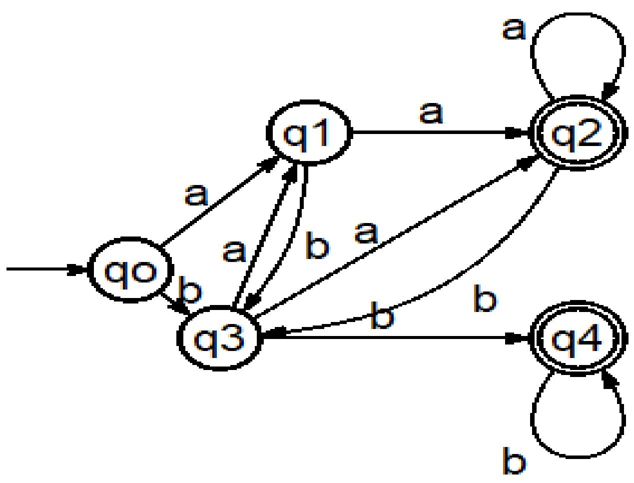
Q11. Find the regular expression to generate the language L={w : |w | mod 3 ≠ 0} on input alphabet ∑={a,b}. (2)

Solution: ((a+b)(a+b)(a+b))* ((a+b)+(a+b)(a+b)) (1+1)

Generate a string which are no multiples of 3. First part generates string of length 3 and part either adds 1 or 2 alphabets so that the string length is not 3X n here n>=1.

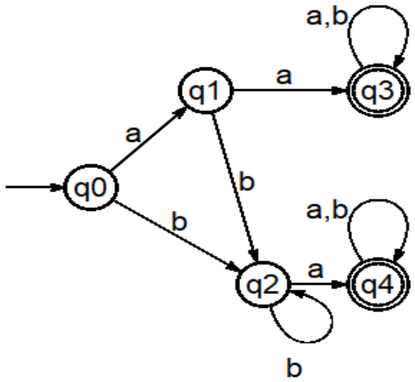
Q12. Design a DFA to accept all the strings ending with aa or bb on ∑ ={a,b}. (2)

Solution:



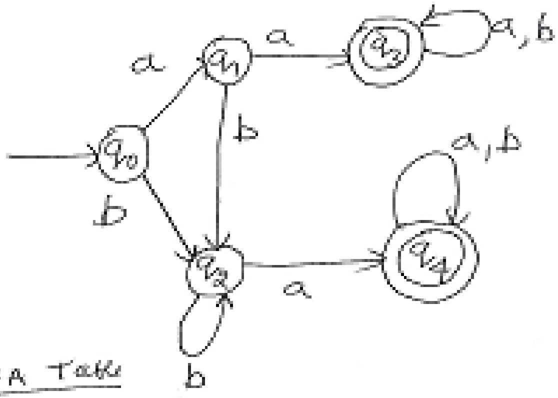
Accepting string ending with aa 1.5 + Accepting string ending with bb 1.5

Q13. Minimize DFA given below using table filling (procedure mark) method. Clearly show all the steps. Draw state diagram for the minimal DFA. (3)



Soluation:

13)



NFA Table

δ_N	a	b
→ q_0	q_1	q_2
q_1	q_3	q_2
q_2	q_4	q_2
q_3	q_3	q_3
q_4	q_4	q_4

(0.5M)

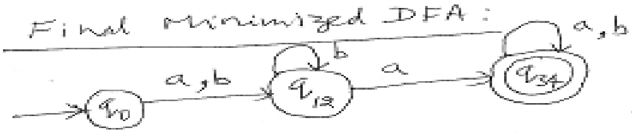
Mark Table:

	q_0	q_1	q_2	q_3	q_4
q_0	.	—	—	—	—
q_1	X	.	—	—	—
q_2	X		.	—	—
q_3	X	X	X	.	—
q_4	X	X	X		.

(1.5 M)

$\delta((q_0, q_1), a) = (q_1, q_3)$ – Mark
$\delta((q_0, q_3), a) = (q_1, q_4)$ – M
$\delta((q_1, q_2), a) = (q_3, q_4)$ – Non Mark
$\delta((q_1, q_2), b) = (q_2, q_2)$ – NM
$\delta((q_3, q_4), a) = (q_3, q_4)$ – NM
$\delta((q_3, q_4), b) = (q_3, q_4)$ – NM
0.5M

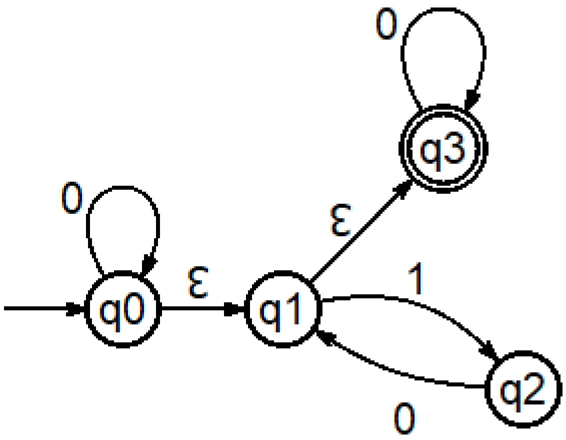
Final Minimized DFA:



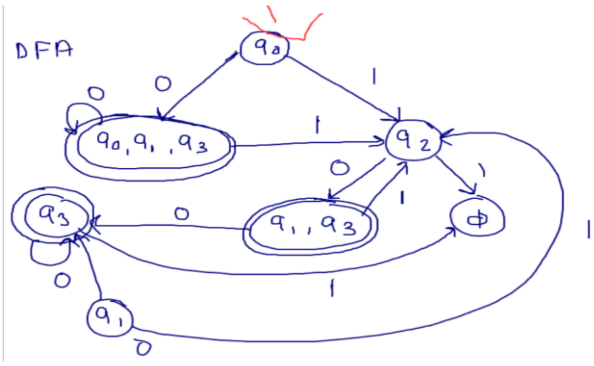
0.5M

Total – 3 Marks

Q14. Convert the below given NFA to its equivalent DFA and represent the same using state diagram, where Ɛ is same as λ (3)



Solution:



1. Elimination of epsilon from NFA -- 1 M

2. Construction of NFA to DFA – 1M

3. Representation of DFA with initial and final sates --1M